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ROUND- 2 Coding Based Question

Total Coding Questions

Topic	Questions
Basic Programming	20
Arrays	25
Strings	25
Pointers	15
Linked List	20
Stack & Queue	20
Trees	20
Graphs	15
Recursion	15
Sorting & Searching	20
Total	195 Coding Question

1. Reverse a Number

Problem

Write a program to reverse a given integer.

Example

Input

12345

Output

54321

C++ Solution

```
#include<iostream>
using namespace std;
```

```
int main()
{
    int n, rev = 0;
    cin >> n;

    while(n != 0)
    {
        rev = rev * 10 + n % 10;
        n /= 10;
    }
}
```

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```
    cout << rev;  
    return 0;  
}
```

Time Complexity: $O(\log n)$

2. Check Palindrome Number

Example

Input

121

Output

Palindrome

```
#include<iostream>  
using namespace std;  
  
int main()  
{  
    int n, temp, rev = 0;  
  
    cin >> n;  
    temp = n;  
  
    while(temp)  
    {  
        rev = rev * 10 + temp % 10;  
        temp /= 10;  
    }  
  
    if(rev == n)  
        cout << "Palindrome";  
    else  
        cout << "Not Palindrome";  
}
```

Time Complexity: $O(\log n)$

3. Check Prime Number

```
#include<iostream>  
using namespace std;
```

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```
int main()
{
    int n;
    bool prime = true;

    cin >> n;

    if(n <= 1)
        prime = false;

    for(int i = 2; i * i <= n; i++)
    {
        if(n % i == 0)
        {
            prime = false;
            break;
        }
    }

    if(prime)
        cout << "Prime";
    else
        cout << "Not Prime";
}
```

Time Complexity: $O(\sqrt{n})$

4. Fibonacci Series

```
#include<iostream>
using namespace std;

int main()
{
    int n;
    cin >> n;

    int a = 0, b = 1;

    for(int i = 0; i < n; i++)
    {
        cout << a << " ";

        int c = a + b;

        a = b;
```

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```
        b = c;
    }
}
```

Time Complexity: $O(n)$

5. Factorial of a Number

```
#include<iostream>
using namespace std;

int main()
{
    int n;
    long long fact = 1;

    cin >> n;

    for(int i = 1; i <= n; i++)
        fact *= i;

    cout << fact;
}
```

Time Complexity: $O(n)$

6. GCD of Two Numbers

```
#include<iostream>
using namespace std;

int main()
{
    int a, b;

    cin >> a >> b;

    while(b)
    {
        int t = b;

        b = a % b;

        a = t;
    }
}
```

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```
    cout << a;
}
```

Time Complexity: $O(\log n)$

7. LCM of Two Numbers

```
#include<iostream>
using namespace std;

int gcd(int a,int b)
{
    while(b)
    {
        int t=b;
        b=a%b;
        a=t;
    }
    return a;
}

int main()
{
    int a,b;

    cin>>a>>b;

    cout<<(a*b)/gcd(a,b);
}
```

8. Armstrong Number

```
#include<iostream>
#include<cmath>
using namespace std;

int main()
{
    int n,temp,sum=0;

    cin>>n;

    temp=n;

    while(temp)
    {
        int d=temp%10;
```

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```
    sum+=pow(d,3);

    temp/=10;
}

if(sum==n)

    cout<<"Armstrong";

else

    cout<<"Not Armstrong";
}
```

9. Sum of Digits

```
#include<iostream>
using namespace std;

int main()
{
    int n,sum=0;

    cin>>n;

    while(n)
    {
        sum+=n%10;

        n/=10;
    }

    cout<<sum;
}
```

10. Count Digits

```
#include<iostream>
using namespace std;

int main()
{
    int n,count=0;

    cin>>n;

    while(n)
```

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```
{  
    count++;  
  
    n/=10;  
}  
  
cout<<count;  
}
```

11. Swap Two Numbers (Without Third Variable)

```
#include<iostream>  
using namespace std;  
  
int main()  
{  
    int a,b;  
  
    cin>>a>>b;  
  
    a=a+b;  
  
    b=a-b;  
  
    a=a-b;  
  
    cout<<a<<" "<<b;  
}
```

12. Largest of Three Numbers

```
#include<iostream>  
using namespace std;  
  
int main()  
{  
    int a,b,c;  
  
    cin>>a>>b>>c;  
  
    if(a>=b && a>=c)  
  
        cout<<a;  
  
    else if(b>=c)  
  
        cout<<b;
```

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```
else  
  
    cout<<c;  
}
```

13. Leap Year

```
#include<iostream>  
using namespace std;  
  
int main()  
{  
    int year;  
  
    cin>>year;  
  
    if((year%400==0)|| (year%4==0 && year%100!=0))  
  
        cout<<"Leap Year";  
  
    else  
  
        cout<<"Not Leap Year";  
}
```

14. Decimal to Binary

```
#include<iostream>  
using namespace std;  
  
int main()  
{  
    int n;  
  
    cin>>n;  
  
    string ans="";  
  
    while(n)  
    {  
        ans=char((n%2)+'0')+ans;  
  
        n/=2;  
    }  
  
    cout<<ans;  
}
```


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15. Binary to Decimal

```
#include<iostream>
#include<cmath>
using namespace std;

int main()
{
    int n,i=0,dec=0;

    cin>>n;

    while(n)
    {
        dec+=(n%10)*pow(2,i);

        i++;

        n/=10;
    }

    cout<<dec;
}
```

16. Reverse a String

```
#include<iostream>
#include<algorithm>
using namespace std;

int main()
{
    string s;

    cin>>s;

    reverse(s.begin(),s.end());

    cout<<s;
}
```

17. Count Vowels

```
#include<iostream>
using namespace std;

int main()
{
```

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```
string s;

getline(cin,s);

int count=0;

for(char c:s)
{
    c=tolower(c);

    if(c=='a'||c=='e'||c=='i'||c=='o'||c=='u')

        count++;
}

cout<<count;
}
```

18. Check Even or Odd

```
#include<iostream>
using namespace std;

int main()
{
    int n;

    cin>>n;

    if(n%2==0)

        cout<<"Even";

    else

        cout<<"Odd";
}
```

19. Power of a Number

```
#include<iostream>
using namespace std;

int main()
{
    int a,b;

    cin>>a>>b;
```

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```
long long ans=1;

while(b--)
    ans*=a;

cout<<ans;
}
```

20. Print Star Pyramid

```
#include<iostream>
using namespace std;

int main()
{
    int n;

    cin>>n;

    for(int i=1;i<=n;i++)
    {
        for(int j=1;j<=n-i;j++)
            cout<<" ";

        for(int j=1;j<=2*i-1;j++)
            cout<<"*";

        cout<<endl;
    }
}
```

Q21. Find the Largest Element in an Array

Problem: Find the largest element in an array.

Example

Input:

5
10 45 7 89 23

Output:

89

C++ Solution

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```
#include<iostream>
using namespace std;
```

```
int main()
{
    int n;
    cin>>n;

    int arr[n];

    for(int i=0;i<n;i++)
        cin>>arr[i];

    int max=arr[0];

    for(int i=1;i<n;i++)
        if(arr[i]>max)
            max=arr[i];

    cout<<max;
}
```

Time Complexity: $O(n)$

Q22. Find the Smallest Element

```
#include<iostream>
using namespace std;
```

```
int main()
{
    int n;
    cin>>n;

    int arr[n];

    for(int i=0;i<n;i++)
        cin>>arr[i];

    int min=arr[0];

    for(int i=1;i<n;i++)
        if(arr[i]<min)
            min=arr[i];

    cout<<min;
}
```

Time Complexity: $O(n)$

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Q23. Find the Second Largest Element

```
#include<iostream>
#include<climits>
using namespace std;

int main()
{
    int n;
    cin>>n;

    int arr[n];

    for(int i=0;i<n;i++)
        cin>>arr[i];

    int first=INT_MIN, second=INT_MIN;

    for(int i=0;i<n;i++)
    {
        if(arr[i]>first)
        {
            second=first;
            first=arr[i];
        }
        else if(arr[i]>second && arr[i]!=first)
        {
            second=arr[i];
        }
    }

    cout<<second;
}
```

Time Complexity: $O(n)$

Q24. Reverse an Array

```
#include<iostream>
using namespace std;
```

```
int main()
{
    int n;
    cin>>n;

    int arr[n];

    for(int i=0;i<n;i++)
```

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```
    cin>>arr[i];

    for(int i=n-1;i>=0;i--)
        cout<<arr[i]<<" ";
}
```

Time Complexity: $O(n)$

Q25. Find Sum of Array Elements

```
#include<iostream>
using namespace std;
```

```
int main()
{
    int n,sum=0;

    cin>>n;

    int arr[n];

    for(int i=0;i<n;i++)
    {
        cin>>arr[i];
        sum+=arr[i];
    }

    cout<<sum;
}
```

Time Complexity: $O(n)$

Q26. Count Even and Odd Numbers

```
#include<iostream>
using namespace std;
```

```
int main()
{
    int n,even=0,odd=0;

    cin>>n;

    int arr[n];

    for(int i=0;i<n;i++)
    {
        cin>>arr[i];
```

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```
        if(arr[i]%2==0)
            even++;
        else
            odd++;
    }

    cout<<"Even = "<<even<<endl;
    cout<<"Odd = "<<odd;
}
```

Q27. Search an Element (Linear Search)

```
#include<iostream>
using namespace std;

int main()
{
    int n,key;

    cin>>n;

    int arr[n];

    for(int i=0;i<n;i++)
        cin>>arr[i];

    cin>>key;

    for(int i=0;i<n;i++)
    {
        if(arr[i]==key)
        {
            cout<<"Found";
            return 0;
        }
    }

    cout<<"Not Found";
}
```

Time Complexity: $O(n)$

Q28. Binary Search

```
#include<iostream>
using namespace std;

int main()
{
    int n,key;
```

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```
cin>>n;

int arr[n];

for(int i=0;i<n;i++)
    cin>>arr[i];

cin>>key;

int low=0,high=n-1;

while(low<=high)
{
    int mid=(low+high)/2;

    if(arr[mid]==key)
    {
        cout<<"Found";
        return 0;
    }

    if(arr[mid]<key)
        low=mid+1;
    else
        high=mid-1;
}

cout<<"Not Found";
}
```

Time Complexity: $O(\log n)$

Q29. Remove Duplicates from Sorted Array

```
#include<iostream>
using namespace std;
```

```
int main()
{
    int n;

    cin>>n;

    int arr[n];

    for(int i=0;i<n;i++)
        cin>>arr[i];

    int j=0;
```


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```
for(int i=1;i<n;i++)
{
    if(arr[i]!=arr[j])
    {
        j++;
        arr[j]=arr[i];
    }
}

cout<<j+1;
}
```

Q30. Move All Zeroes to End

```
#include<iostream>
using namespace std;

int main()
{
    int n;

    cin>>n;

    int arr[n];

    for(int i=0;i<n;i++)
        cin>>arr[i];

    int index=0;

    for(int i=0;i<n;i++)
    {
        if(arr[i]!=0)
            swap(arr[index++],arr[i]);
    }

    for(int i=0;i<n;i++)
        cout<<arr[i]<<" ";
}
```

Time Complexity: $O(n)$

Q31. Left Rotate Array by One Position

```
#include<iostream>
using namespace std;

int main()
{
    int n;
```

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```
cin>>n;

int arr[n];

for(int i=0;i<n;i++)
    cin>>arr[i];

int first=arr[0];

for(int i=0;i<n-1;i++)
    arr[i]=arr[i+1];

arr[n-1]=first;

for(int i=0;i<n;i++)
    cout<<arr[i]<<" ";
}
```

Q32. Right Rotate Array by One Position

```
#include<iostream>
using namespace std;

int main()
{
    int n;

    cin>>n;

    int arr[n];

    for(int i=0;i<n;i++)
        cin>>arr[i];

    int last=arr[n-1];

    for(int i=n-1;i>0;i--)
        arr[i]=arr[i-1];

    arr[0]=last;

    for(int i=0;i<n;i++)
        cout<<arr[i]<<" ";
}
```

Q33. Check if Array is Sorted

```
#include<iostream>
using namespace std;

int main()
```

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```
{
    int n;

    cin>>n;

    int arr[n];

    for(int i=0;i<n;i++)
        cin>>arr[i];

    for(int i=0;i<n-1;i++)
    {
        if(arr[i]>arr[i+1])
        {
            cout<<"Not Sorted";
            return 0;
        }
    }

    cout<<"Sorted";
}
```

Q34. Find Missing Number (1 to n)

```
#include<iostream>
using namespace std;

int main()
{
    int n;

    cin>>n;

    int arr[n-1];

    int sum=0;

    for(int i=0;i<n-1;i++)
    {
        cin>>arr[i];
        sum+=arr[i];
    }

    cout<<(n*(n+1))/2-sum;
}
```

Q35. Find Frequency of an Element

```
#include<iostream>
using namespace std;
```

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```
int main()
{
    int n,key,count=0;

    cin>>n;

    int arr[n];

    for(int i=0;i<n;i++)
        cin>>arr[i];

    cin>>key;

    for(int i=0;i<n;i++)
        if(arr[i]==key)
            count++;

    cout<<count;
}
```

Q36. Two Sum

Problem: Find two indices such that their corresponding elements add up to the target.

Example

Input:

```
6
2 7 11 15 3 6
9
```

Output:

```
0 1
```

C++ Solution

```
#include<iostream>
#include<unordered_map>
using namespace std;

int main()
{
    int n,target;
    cin>>n;

    int arr[n];
```

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```
for(int i=0;i<n;i++)
    cin>>arr[i];

cin>>target;

unordered_map<int,int> mp;

for(int i=0;i<n;i++)
{
    int diff=target-arr[i];

    if(mp.find(diff)!=mp.end())
    {
        cout<<mp[diff]<<" "<<i;
        return 0;
    }

    mp[arr[i]]=i;
}

cout<<"No Pair Found";
}
```

Time Complexity: $O(n)$

Q37. Kadane's Algorithm (Maximum Subarray Sum)

Problem: Find the maximum sum of a contiguous subarray.

Example

Input:

8
-2 -3 4 -1 -2 1 5 -3

Output:

7

C++ Solution

```
#include<iostream>
#include<climits>
using namespace std;
```

```
int main()
{
```

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```
int n;
cin>>n;

int arr[n];

for(int i=0;i<n;i++)
    cin>>arr[i];

int maxSum=INT_MIN;
int current=0;

for(int i=0;i<n;i++)
{
    current+=arr[i];

    if(current>maxSum)
        maxSum=current;

    if(current<0)
        current=0;
}

cout<<maxSum;
}
```

Time Complexity: $O(n)$

Q38. Majority Element

Problem: Find the element occurring more than $n/2$ times.

Example

Input:

```
7
2 2 1 2 3 2 2
```

Output:

```
2
```

C++ Solution (Moore's Voting Algorithm)

```
#include<iostream>
using namespace std;
```

```
int main()
```

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```
{
    int n;
    cin>>n;

    int arr[n];

    for(int i=0;i<n;i++)
        cin>>arr[i];

    int candidate=arr[0];
    int count=1;

    for(int i=1;i<n;i++)
    {
        if(arr[i]==candidate)
            count++;
        else
            count--;

        if(count==0)
        {
            candidate=arr[i];
            count=1;
        }
    }

    cout<<candidate;
}
```

Time Complexity: $O(n)$

Q39. Merge Two Sorted Arrays

Example

Input:

```
3
1 3 5
3
2 4 6
```

Output:

```
1 2 3 4 5 6
```

C++ Solution

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```
#include<iostream>
using namespace std;

int main()
{
    int n,m;

    cin>>n;

    int a[n];

    for(int i=0;i<n;i++)
        cin>>a[i];

    cin>>m;

    int b[m];

    for(int i=0;i<m;i++)
        cin>>b[i];

    int i=0,j=0;

    while(i<n && j<m)
    {
        if(a[i]<b[j])
            cout<<a[i++]<<" ";
        else
            cout<<b[j++]<<" ";
    }

    while(i<n)
        cout<<a[i++]<<" ";

    while(j<m)
        cout<<b[j++]<<" ";
}
```

Time Complexity: $O(n+m)$

Q40. Prefix Sum Array

Problem: Create the prefix sum array.

Example

Input:

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5
2 4 6 8 10

Output:

2 6 12 20 30

C++ Solution

```
#include<iostream>
using namespace std;

int main()
{
    int n;
    cin>>n;

    int arr[n];

    for(int i=0;i<n;i++)
        cin>>arr[i];

    for(int i=1;i<n;i++)
        arr[i]+=arr[i-1];

    for(int i=0;i<n;i++)
        cout<<arr[i]<<" ";
}
```

Time Complexity: $O(n)$

Q41. Best Time to Buy and Sell Stock

Problem: Find the maximum profit from one buy and one sell.

Example

Input:

6
7 1 5 3 6 4

Output:

5

C++ Solution

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```
#include<iostream>
#include<climits>
using namespace std;

int main()
{
    int n;
    cin>>n;

    int price[n];

    for(int i=0;i<n;i++)
        cin>>price[i];

    int minPrice=INT_MAX;
    int profit=0;

    for(int i=0;i<n;i++)
    {
        if(price[i]<minPrice)
            minPrice=price[i];

        if(price[i]-minPrice>profit)
            profit=price[i]-minPrice;
    }

    cout<<profit;
}
```

Time Complexity: $O(n)$

Q42. Intersection of Two Arrays

Example

Input:

```
5
1 2 3 4 5
5
3 4 5 6 7
```

Output:

```
3 4 5
```

C++ Solution

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```
#include<iostream>
#include<set>
using namespace std;

int main()
{
    int n,m;

    cin>>n;

    set<int> s;

    for(int i=0;i<n;i++)
    {
        int x;
        cin>>x;
        s.insert(x);
    }

    cin>>m;

    for(int i=0;i<m;i++)
    {
        int x;
        cin>>x;

        if(s.count(x))
            cout<<x<<" ";
    }
}
```

Time Complexity: $O(n \log n)$

Q43. Leaders in an Array

Problem: Print elements greater than all elements to their right.

Example

Input:

6
16 17 4 3 5 2

Output:

17 5 2

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C++ Solution

```
#include<iostream>
#include<vector>
using namespace std;

int main()
{
    int n;
    cin>>n;

    int arr[n];

    for(int i=0;i<n;i++)
        cin>>arr[i];

    vector<int> ans;

    int leader=arr[n-1];

    ans.push_back(leader);

    for(int i=n-2;i>=0;i--)
    {
        if(arr[i]>leader)
        {
            leader=arr[i];
            ans.push_back(leader);
        }
    }

    for(int i=ans.size()-1;i>=0;i--)
        cout<<ans[i]<<" ";
}
```

Time Complexity: $O(n)$

Q44. Maximum Product Subarray

Problem: Find the maximum product of a contiguous subarray.

Example

Input:

4
2 3 -2 4

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Output:

6

C++ Solution

```
#include<iostream>
#include<algorithm>
using namespace std;

int main()
{
    int n;
    cin>>n;

    int arr[n];

    for(int i=0;i<n;i++)
        cin>>arr[i];

    int maxProd=arr[0];
    int minProd=arr[0];
    int ans=arr[0];

    for(int i=1;i<n;i++)
    {
        if(arr[i]<0)
            swap(maxProd,minProd);

        maxProd=max(arr[i],maxProd*arr[i]);
        minProd=min(arr[i],minProd*arr[i]);

        ans=max(ans,maxProd);
    }

    cout<<ans;
}
```

Time Complexity: $O(n)$

[Q45. Rotate Array by K Positions](#)

Example

Input:

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5
1 2 3 4 5
2

Output:

4 5 1 2 3

C++ Solution

```
#include<iostream>
#include<algorithm>
using namespace std;

int main()
{
    int n,k;
    cin>>n;

    int arr[n];

    for(int i=0;i<n;i++)
        cin>>arr[i];

    cin>>k;

    k%=n;

    reverse(arr,arr+n);
    reverse(arr,arr+k);
    reverse(arr+k,arr+n);

    for(int i=0;i<n;i++)
        cout<<arr[i]<<" ";
}
```

Q46. Reverse a String

Problem: Reverse a given string.

Example

Input:

Hello

Output:

olleH

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C++ Solution

```
#include<iostream>
#include<algorithm>
using namespace std;

int main()
{
    string str;
    cin>>str;

    reverse(str.begin(),str.end());

    cout<<str;
}
```

Time Complexity: $O(n)$

Q47. Check Palindrome String

Problem: Check whether a string is a palindrome.

Example

Input:

madam

Output:

Palindrome

C++ Solution

```
#include<iostream>
using namespace std;

int main()
{
    string str;
    cin>>str;

    int i=0,j=str.length()-1;

    while(i<j)
    {
        if(str[i]!=str[j])
        {

```

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```
        cout<<"Not Palindrome";
        return 0;
    }
    i++;
    j--;
}

cout<<"Palindrome";
}
```

Time Complexity: $O(n)$

Q48. Check Anagram

Problem: Check whether two strings are anagrams.

Example

Input

listen
silent

Output

Anagram

C++ Solution

```
#include<iostream>
#include<algorithm>
using namespace std;

int main()
{
    string a,b;

    cin>>a>>b;

    sort(a.begin(),a.end());
    sort(b.begin(),b.end());

    if(a==b)
        cout<<"Anagram";
    else
        cout<<"Not Anagram";
}
```


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Time Complexity: $O(n \log n)$

Q49. Count Vowels and Consonants

Example

Input

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Output

Vowels = 3

Consonants = 8

C++ Solution

```
#include<iostream>
using namespace std;

int main()
{
    string str;

    cin>>str;

    int vowels=0,consonants=0;

    for(char ch:str)
    {
        ch=tolower(ch);

        if(ch=='a' || ch=='e' || ch=='i' || ch=='o' || ch=='u')
            vowels++;
        else if(isalpha(ch))
            consonants++;
    }

    cout<<"Vowels = "<<vowels<<endl;
    cout<<"Consonants = "<<consonants;
}
```

Time Complexity: $O(n)$

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Q50. Count Frequency of Each Character

Example

Input

apple

Output

a : 1

p : 2

l : 1

e : 1

C++ Solution

```
#include<iostream>
#include<map>
using namespace std;

int main()
{
    string str;

    cin>>str;

    map<char,int> freq;

    for(char ch:str)
        freq[ch]++;

    for(auto x:freq)
        cout<<x.first<<" : "<<x.second<<endl;
}
```

Time Complexity: $O(n \log n)$

Q51. Remove Spaces from a String

Example

Input

Hello World Program

Output

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HelloWorldProgram

C++ Solution

```
#include<iostream>
using namespace std;
```

```
int main()
{
    string str;

    getline(cin,str);

    for(char ch:str)
    {
        if(ch!=' ')
            cout<<ch;
    }
}
```

Time Complexity: $O(n)$

Q52. Convert Lowercase to Uppercase

Example

Input

welcome

Output

WELCOME

C++ Solution

```
#include<iostream>
#include<algorithm>
using namespace std;
```

```
int main()
{
    string str;

    cin>>str;

    for(char &ch:str)
        ch=toupper(ch);
}
```

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```
    cout<<str;  
}
```

Time Complexity: $O(n)$

Q53. Find Length of String Without Using length()

Example

Input

Computer

Output

8

C++ Solution

```
#include<iostream>  
using namespace std;  
  
int main()  
{  
    char str[100];  
  
    cin>>str;  
  
    int count=0;  
  
    while(str[count]!='\0')  
        count++;  
  
    cout<<count;  
}
```

Time Complexity: $O(n)$

Q54. Reverse Words in a Sentence

Example

Input

I Love Coding

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Output

Coding Love I

C++ Solution

```
#include<iostream>
#include<sstream>
#include<vector>
using namespace std;

int main()
{
    string str;

    getline(cin,str);

    stringstream ss(str);

    vector<string> words;

    string word;

    while(ss>>word)
        words.push_back(word);

    for(int i=words.size()-1;i>=0;i--)
        cout<<words[i]<<" ";
}
```

Time Complexity: $O(n)$

Q55. Find the First Non-Repeating Character

Example

Input

aabbcdde

Output

c

C++ Solution

```
#include<iostream>
#include<unordered_map>
```

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```
using namespace std;

int main()
{
    string str;

    cin>>str;

    unordered_map<char,int> freq;

    for(char ch:str)
        freq[ch]++;

    for(char ch:str)
    {
        if(freq[ch]==1)
        {
            cout<<ch;
            return 0;
        }
    }

    cout<<"No Unique Character";
}
```

Q56. Longest Common Prefix

Problem: Find the longest common prefix among multiple strings.

Example

Input

```
3
flower
flow
flight
```

Output

```
fl
```

C++ Solution

```
#include<iostream>
#include<vector>
#include<algorithm>
using namespace std;
```

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```
int main()
{
    int n;
    cin>>n;

    vector<string> str(n);

    for(int i=0;i<n;i++)
        cin>>str[i];

    sort(str.begin(),str.end());

    string first=str[0];
    string last=str[n-1];

    string ans="";

    for(int i=0;i<min(first.length(),last.length());i++)
    {
        if(first[i]==last[i])
            ans+=first[i];
        else
            break;
    }

    cout<<ans;
}
```

Time Complexity: $O(n \log n)$

Q57. Valid Parentheses

Problem: Check whether the parentheses are balanced.

Example

Input

({[]})

Output

Balanced

C++ Solution

```
#include<iostream>
#include<stack>
```

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```
using namespace std;

int main()
{
    string str;
    cin>>str;

    stack<char> st;

    for(char ch:str)
    {
        if(ch=='('||ch=='{'||ch=='[')
            st.push(ch);

        else
        {
            if(st.empty())
            {
                cout<<"Not Balanced";
                return 0;
            }

            if((ch==')'&&st.top()!='(')||
               (ch=='}'&&st.top()!='{'||
               (ch==']'&&st.top()!='['))
            {
                cout<<"Not Balanced";
                return 0;
            }

            st.pop();
        }
    }

    if(st.empty())
        cout<<"Balanced";
    else
        cout<<"Not Balanced";
}
```

Time Complexity: $O(n)$

Q58. String Compression

Problem: Compress a string by counting consecutive characters.

Example

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Input

aaabbccccc

Output

a3b2c4

C++ Solution

```
#include<iostream>
using namespace std;

int main()
{
    string str;
    cin>>str;

    int count=1;

    for(int i=0;i<str.length();i++)
    {
        while(i<str.length()-1 && str[i]==str[i+1])
        {
            count++;
            i++;
        }

        cout<<str[i]<<count;
        count=1;
    }
}
```

Time Complexity: $O(n)$

Q59. Check String Rotation

Problem: Check whether one string is a rotation of another.

Example

Input

abcd
cdab

Output

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Yes

C++ Solution

```
#include<iostream>
using namespace std;

int main()
{
    string s1,s2;

    cin>>s1>>s2;

    if(s1.length()!=s2.length())
    {
        cout<<"No";
        return 0;
    }

    string temp=s1+s1;

    if(temp.find(s2)!=string::npos)
        cout<<"Yes";
    else
        cout<<"No";
}
```

Time Complexity: O(n)

Q60. Remove Duplicate Characters

Problem: Remove duplicate characters while preserving order.

Example

Input

programming

Output

progamin

C++ Solution

```
#include<iostream>
#include<set>
using namespace std;
```

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```
int main()
{
    string str;
    cin>>str;

    set<char> s;

    for(char ch:str)
    {
        if(s.find(ch)==s.end())
        {
            cout<<ch;
            s.insert(ch);
        }
    }
}
```

Time Complexity: $O(n \log n)$

Q61. Longest Palindromic Substring (Brute Force)

Problem: Find the longest palindromic substring.

Example

Input

babad

Output

bab

C++ Solution

```
#include<iostream>
using namespace std;

bool palindrome(string s,int l,int r)
{
    while(l<r)
    {
        if(s[l]!=s[r])
            return false;
        l++;
        r--;
    }
}
```

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```
        return true;
    }

    int main()
    {
        string s;
        cin>>s;

        string ans="";

        for(int i=0;i<s.length();i++)
        {
            for(int j=i;j<s.length();j++)
            {
                if(palindrome(s,i,j) && j-i+1>ans.length())
                    ans=s.substr(i,j-i+1);
            }
        }

        cout<<ans;
    }
```

Time Complexity: $O(n^3)$

Q62. Check Isomorphic Strings

Problem: Determine whether two strings are isomorphic.

Example

Input

egg
add

Output

Yes

C++ Solution

```
#include<iostream>
#include<unordered_map>
using namespace std;

int main()
{
    string s,t;
```

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```
cin>>s>>t;

unordered_map<char,char> m1,m2;

for(int i=0;i<s.length();i++)
{
    if((m1.count(s[i]) && m1[s[i]]!=t[i]) ||
        (m2.count(t[i]) && m2[t[i]]!=s[i]))
    {
        cout<<"No";
        return 0;
    }

    m1[s[i]]=t[i];
    m2[t[i]]=s[i];
}

cout<<"Yes";
}
```

Time Complexity: $O(n)$

Q63. Find a Substring

Problem: Find the first occurrence of a substring.

Example

Input

programming
gram

Output

3

C++ Solution

```
#include<iostream>
using namespace std;
```

```
int main()
{
    string str,sub;

    cin>>str>>sub;
```

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```
    cout<<str.find(sub);  
}
```

Time Complexity: $O(n)$

Q64. Roman to Integer

Example

Input

XIV

Output

14

C++ Solution

```
#include<iostream>  
#include<unordered_map>  
using namespace std;  
  
int main()  
{  
    string s;  
    cin>>s;  
  
    unordered_map<char,int> mp;  
  
    mp['I']=1;  
    mp['V']=5;  
    mp['X']=10;  
    mp['L']=50;  
    mp['C']=100;  
    mp['D']=500;  
    mp['M']=1000;  
  
    int ans=0;  
  
    for(int i=0;i<s.length();i++)  
    {  
        if(i<s.length()-1 && mp[s[i]]<mp[s[i+1]])  
            ans-=mp[s[i]];  
        else  
            ans+=mp[s[i]];  
    }  
}
```

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```
    cout<<ans;  
}
```

Time Complexity: $O(n)$

Q65. Integer to Roman

Example

Input

58

Output

LVIII

C++ Solution

```
#include<iostream>  
using namespace std;  
  
int main()  
{  
    int num;  
    cin>>num;  
  
    string roman[]={ "M","CM","D","CD","C","XC","L","XL","X","IX","V","IV","I"};  
    int value[]={1000,900,500,400,100,90,50,40,10,9,5,4,1};  
  
    for(int i=0;i<13;i++)  
    {  
        while(num>=value[i])  
        {  
            cout<<roman[i];  
            num-=value[i];  
        }  
    }  
}
```

Q66. Edit Distance (Levenshtein Distance)

Problem: Find the minimum number of operations (Insert, Delete, Replace) required to convert one string into another.

Example

Input

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horse
ros

Output

3

C++ Solution

```
#include<iostream>
#include<vector>
#include<algorithm>
using namespace std;

int main()
{
    string s1,s2;
    cin>>s1>>s2;

    int m=s1.length();
    int n=s2.length();

    vector<vector<int>>> dp(m+1,vector<int>(n+1));

    for(int i=0;i<=m;i++)
        dp[i][0]=i;

    for(int j=0;j<=n;j++)
        dp[0][j]=j;

    for(int i=1;i<=m;i++)
    {
        for(int j=1;j<=n;j++)
        {
            if(s1[i-1]==s2[j-1])
                dp[i][j]=dp[i-1][j-1];
            else
                dp[i][j]=1+min({dp[i-1][j],dp[i][j-1],dp[i-1][j-1]});
        }
    }

    cout<<dp[m][n];
}
```

Time Complexity: $O(m \times n)$

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Q67. Longest Substring Without Repeating Characters

Problem: Find the length of the longest substring without repeating characters.

Example

Input

abcabcbb

Output

3

C++ Solution

```
#include<iostream>
#include<unordered_map>
using namespace std;

int main()
{
    string s;
    cin>>s;

    unordered_map<char,int> mp;

    int start=0;
    int ans=0;

    for(int end=0;end<s.length();end++)
    {
        if(mp.count(s[end]))
            start=max(start,mp[s[end]]+1);

        mp[s[end]]=end;

        ans=max(ans,end-start+1);
    }

    cout<<ans;
}
```

Time Complexity: O(n)

Q68. Implement atoi() (String to Integer)

Problem: Convert a numeric string into an integer.

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Example

Input

12345

Output

12345

C++ Solution

```
#include<iostream>
using namespace std;

int main()
{
    string s;
    cin>>s;

    int num=0;

    for(char ch:s)
    {
        num=num*10+(ch-'0');
    }

    cout<<num;
}
```

Time Complexity: $O(n)$

Q69. Implement strStr() (Pattern Matching)

Problem: Find the first occurrence of a pattern in a string.

Example

Input

helloworld
world

Output

5

C++ Solution

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```
#include<iostream>
using namespace std;

int main()
{
    string text,pattern;

    cin>>text>>pattern;

    int pos=text.find(pattern);

    if(pos==-1)
        cout<<"Not Found";
    else
        cout<<pos;
}
```

Time Complexity: $O(n \times m)$

Q70. Count Occurrences of a Pattern

Problem: Count how many times a pattern appears in a string.

Example

Input

abababab
ab

Output

4

C++ Solution

```
#include<iostream>
using namespace std;

int main()
{
    string text,pattern;

    cin>>text>>pattern;

    int count=0;

    for(int i=0;i<=text.length()-pattern.length();i++)
```

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```
{
    if(text.substr(i,pattern.length())==pattern)
        count++;
}

cout<<count;
}
```

Time Complexity: $O(n \times m)$

Q71. Swap Two Numbers Using Pointers

Problem: Swap two numbers using pointers.

Example

Input

10 20

Output

20 10

C++ Solution

```
#include<iostream>
using namespace std;

void swapNum(int *a,int *b)
{
    int temp=*a;
    *a=*b;
    *b=temp;
}

int main()
{
    int a,b;

    cin>>a>>b;

    swapNum(&a,&b);

    cout<<a<<" "<<b;
}
```

Time Complexity: $O(1)$

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Q72. Find Sum of Array Using Pointers

Example

Input

5
1 2 3 4 5

Output

15

C++ Solution

```
#include<iostream>
using namespace std;
```

```
int main()
{
    int n,sum=0;

    cin>>n;

    int arr[n];

    for(int i=0;i<n;i++)
        cin>>arr[i];

    int *ptr=arr;

    for(int i=0;i<n;i++)
        sum+=*(ptr+i);

    cout<<sum;
}
```

Time Complexity: $O(n)$

Q73. Find Largest Element Using Pointer

Example

Input

5
12 55 31 98 44

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Output

98

C++ Solution

```
#include<iostream>
using namespace std;
```

```
int main()
{
    int n;

    cin>>n;

    int arr[n];

    for(int i=0;i<n;i++)
        cin>>arr[i];

    int *ptr=arr;

    int max=*ptr;

    for(int i=1;i<n;i++)
    {
        if(*(ptr+i)>max)
            max=*(ptr+i);
    }

    cout<<max;
}
```

Time Complexity: $O(n)$

Q74. Reverse Array Using Pointers

Example

Input

5
1 2 3 4 5

Output

5 4 3 2 1

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C++ Solution

```
#include<iostream>
using namespace std;

int main()
{
    int n;

    cin>>n;

    int arr[n];

    for(int i=0;i<n;i++)
        cin>>arr[i];

    int *start=arr;
    int *end=arr+n-1;

    while(start<end)
    {
        int temp=*start;
        *start=*end;
        *end=temp;

        start++;
        end--;
    }

    for(int i=0;i<n;i++)
        cout<<arr[i]<<" ";
}
```

Time Complexity: $O(n)$

Q75. Pointer Arithmetic

Problem: Print all elements using pointer arithmetic.

Example

Input

4
5 10 15 20

Output

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5 10 15 20

C++ Solution

```
#include<iostream>
using namespace std;

int main()
{
    int n;

    cin>>n;

    int arr[n];

    for(int i=0;i<n;i++)
        cin>>arr[i];

    int *ptr=arr;

    for(int i=0;i<n;i++)
        cout<<*(ptr+i)<<" ";
}
```

Time Complexity: $O(n)$

Q76. Pointer to Pointer (Double Pointer)

Example

Input

25

Output

25

C++ Solution

```
#include<iostream>
using namespace std;

int main()
{
    int x;

    cin>>x;
```


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```
int *p=&x;

int **q=&p;

cout<<**q;
}
```

Time Complexity: $O(1)$

Q77. Dynamically Allocate Memory

Example

Input

5

Output

Memory Allocated

C++ Solution

```
#include<iostream>
using namespace std;

int main()
{
    int n;

    cin>>n;

    int *arr=new int[n];

    if(arr!=NULL)
        cout<<"Memory Allocated";

    delete[] arr;
}
```

Time Complexity: $O(1)$

Q78. Find Length of String Using Pointer

Example

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Input

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Output

11

C++ Solution

```
#include<iostream>
using namespace std;
```

```
int main()
{
    char str[100];

    cin>>str;

    char *ptr=str;

    int count=0;

    while(*ptr!='\0')
    {
        count++;
        ptr++;
    }

    cout<<count;
}
```

Time Complexity: $O(n)$

Q79. Copy One String to Another Using Pointer

Example

Input

Hello

Output

Hello

C++ Solution

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```
#include<iostream>
using namespace std;

int main()
{
    char str1[100],str2[100];

    cin>>str1;

    char *p1=str1;
    char *p2=str2;

    while(*p1!='\0')
    {
        *p2=*p1;
        p1++;
        p2++;
    }

    *p2='\0';

    cout<<str2;
}
```

Time Complexity: $O(n)$

Q80. Compare Two Strings Using Pointer

Example

Input

Hello
Hello

Output

Equal

C++ Solution

```
#include<iostream>
using namespace std;

int main()
{
    char s1[100],s2[100];
```

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```
cin>>s1>>s2;

char *p1=s1;
char *p2=s2;

while(*p1&&*p2)
{
    if(*p1!=*p2)
    {
        cout<<"Not Equal";
        return 0;
    }

    p1++;
    p2++;
}

if(*p1==*p2)
    cout<<"Equal";
else
    cout<<"Not Equal";
}
```

Q81. Function Pointer

Problem: Use a function pointer to perform addition.

Example

Input

10 20

Output

30

C++ Solution

```
#include<iostream>
using namespace std;
```

```
int add(int a,int b)
{
    return a+b;
}
```

```
int main()
{
```

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```
int a,b;
cin>>a>>b;

int (*ptr)(int,int);

ptr=add;

cout<<ptr(a,b);

return 0;
}
```

Time Complexity: $O(1)$

Q82. Void Pointer

Problem: Store an integer using a void pointer and print it.

Example

Input

25

Output

25

C++ Solution

```
#include<iostream>
using namespace std;

int main()
{
    int x;
    cin>>x;

    void *ptr;

    ptr=&x;

    cout<<*(int*)ptr;

    return 0;
}
```

Time Complexity: $O(1)$

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Q83. Pointer to Structure

Problem: Access structure members using a pointer.

Example

Input

Rahul
101

Output

Name : Rahul
ID : 101

C++ Solution

```
#include<iostream>
using namespace std;

struct Student
{
    string name;
    int id;
};

int main()
{
    Student s;

    cin>>s.name>>s.id;

    Student *ptr=&s;

    cout<<"Name : "<<ptr->name<<endl;
    cout<<"ID : "<<ptr->id;

    return 0;
}
```

Time Complexity: O(1)

Q84. Dangling Pointer

Problem: Demonstrate how to avoid a dangling pointer.

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Example

Output

Pointer is NULL

C++ Solution

```
#include<iostream>
using namespace std;

int main()
{
    int *ptr=new int(100);

    delete ptr;

    ptr=NULL;

    if(ptr==NULL)
        cout<<"Pointer is NULL";

    return 0;
}
```

Time Complexity: O(1)

Interview Tip: A dangling pointer points to memory that has already been deallocated. Always assign nullptr (or NULL in older code) after deleting dynamically allocated memory.

Q85. Dynamic Memory Allocation (malloc, calloc, realloc)

Problem: Demonstrate dynamic memory allocation in C.

Example

Input

5

Output

10 20 30 40 50

C Solution

```
#include<stdio.h>
#include<stdlib.h>
```

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```
int main()
{
    int n,i;

    scanf("%d",&n);

    int *arr=(int*)malloc(n*sizeof(int));

    for(i=0;i<n;i++)
        arr[i]=(i+1)*10;

    for(i=0;i<n;i++)
        printf("%d ",arr[i]);

    free(arr);

    return 0;
}
```

Time Complexity: $O(n)$

✦ Frequently Asked Interview Follow-ups

Q. Difference between malloc() and calloc()?

malloc()	calloc()
Allocates one block of memory	Allocates multiple blocks
Memory is not initialized	Memory is initialized to 0
Faster	Slightly slower due to initialization

Q. What does realloc() do?

- Changes the size of previously allocated memory.
- Existing data is preserved (up to the minimum of old and new sizes).

Example:

```
arr = (int*)realloc(arr, 10 * sizeof(int));
```


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Linked List Coding Questions (Q86–Q105)

Q86. Insert Node at Beginning

Example

Input

10 20 30

Insert : 5

Output

5 10 20 30

C++ Solution

```
#include<iostream>
using namespace std;

struct Node
{
    int data;
    Node* next;
};

void print(Node* head)
{
    while(head)
    {
        cout<<head->data<<" ";
        head=head->next;
    }
}

int main()
{
    Node *head=new Node{10,NULL};
    head->next=new Node{20,NULL};
    head->next->next=new Node{30,NULL};

    Node *temp=new Node{5,head};

    head=temp;

    print(head);
}
```

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Time Complexity: $O(1)$

Q87. Insert Node at End

```
#include<iostream>
using namespace std;

struct Node
{
    int data;
    Node *next;
};

int main()
{
    Node *head=new Node{10,NULL};
    head->next=new Node{20,NULL};

    Node *temp=head;

    while(temp->next)
        temp=temp->next;

    temp->next=new Node{30,NULL};

    temp=head;

    while(temp)
    {
        cout<<temp->data<<" ";
        temp=temp->next;
    }
}
```

Time Complexity: $O(n)$

Q88. Insert at Given Position

```
#include<iostream>
using namespace std;

struct Node
{
    int data;
    Node *next;
};

int main()
{
    Node *head=new Node{10,NULL};
    head->next=new Node{20,NULL};
    head->next->next=new Node{40,NULL};
}
```

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```
int pos=3;

Node *temp=head;

for(int i=1;i<pos-1;i++)
    temp=temp->next;

Node *node=new Node{30,temp->next};

temp->next=node;

temp=head;

while(temp)
{
    cout<<temp->data<<" ";
    temp=temp->next;
}
}
```

Time Complexity: $O(n)$

Q89. Delete First Node

```
#include<iostream>
using namespace std;

struct Node
{
    int data;
    Node *next;
};

int main()
{
    Node *head=new Node{10,NULL};
    head->next=new Node{20,NULL};

    Node *temp=head;

    head=head->next;

    delete temp;

    while(head)
    {
        cout<<head->data<<" ";
        head=head->next;
    }
}
```

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Time Complexity: $O(1)$

Q90. Delete Last Node

```
#include<iostream>
using namespace std;

struct Node
{
    int data;
    Node *next;
};

int main()
{
    Node *head=new Node{10,NULL};
    head->next=new Node{20,NULL};
    head->next->next=new Node{30,NULL};

    Node *temp=head;

    while(temp->next->next)
        temp=temp->next;

    delete temp->next;

    temp->next=NULL;

    temp=head;

    while(temp)
    {
        cout<<temp->data<<" ";
        temp=temp->next;
    }
}
```

Time Complexity: $O(n)$

Q91. Search an Element

```
#include<iostream>
using namespace std;

struct Node
{
    int data;
    Node *next;
};

int main()
{
```

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```
Node *head=new Node{10,NULL};
head->next=new Node{20,NULL};
head->next->next=new Node{30,NULL};

int key=20;

while(head)
{
    if(head->data==key)
    {
        cout<<"Found";
        return 0;
    }

    head=head->next;
}

cout<<"Not Found";
}
```

Time Complexity: $O(n)$

Q92. Count Nodes

```
#include<iostream>
using namespace std;

struct Node
{
    int data;
    Node *next;
};

int main()
{
    Node *head=new Node{10,NULL};
    head->next=new Node{20,NULL};
    head->next->next=new Node{30,NULL};

    int count=0;

    while(head)
    {
        count++;
        head=head->next;
    }

    cout<<count;
}
```

Time Complexity: $O(n)$

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Q93. Reverse Linked List ★ ★ ★

```
#include<iostream>
using namespace std;

struct Node
{
    int data;
    Node *next;
};

int main()
{
    Node *head=new Node{10,NULL};
    head->next=new Node{20,NULL};
    head->next->next=new Node{30,NULL};

    Node *prev=NULL;
    Node *curr=head;

    while(curr)
    {
        Node *next=curr->next;

        curr->next=prev;

        prev=curr;

        curr=next;
    }

    while(prev)
    {
        cout<<prev->data<<" ";

        prev=prev->next;
    }
}
```

Time Complexity: $O(n)$

Q94. Find Middle Node ★ ★ ★

```
#include<iostream>
using namespace std;

struct Node
{

```

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```
int data;
Node *next;
};

int main()
{
    Node *head=new Node{10,NULL};
    head->next=new Node{20,NULL};
    head->next->next=new Node{30,NULL};
    head->next->next->next=new Node{40,NULL};
    head->next->next->next->next=new Node{50,NULL};

    Node *slow=head;
    Node *fast=head;

    while(fast && fast->next)
    {
        slow=slow->next;

        fast=fast->next->next;
    }

    cout<<slow->data;
}
```

Time Complexity: $O(n)$

Q95. Find Nth Node from End ★ ★ ★

```
#include<iostream>
using namespace std;

struct Node
{
    int data;
    Node *next;
};

int main()
{
    Node *head=new Node{10,NULL};
    head->next=new Node{20,NULL};
    head->next->next=new Node{30,NULL};
    head->next->next->next=new Node{40,NULL};

    int n=2;

    Node *first=head;
    Node *second=head;

    for(int i=0;i<n;i++)
```

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```
first=first->next;

while(first)
{
    first=first->next;
    second=second->next;
}

cout<<second->data;
}
```

Time Complexity: $O(n)$

Q96. Detect Loop in Linked List (Floyd's Cycle Detection)

Problem: Detect whether a linked list contains a loop.

Output

Loop Found

[C++ Solution](#)

```
#include<iostream>
using namespace std;

struct Node
{
    int data;
    Node *next;
};

bool detectLoop(Node *head)
{
    Node *slow=head;
    Node *fast=head;

    while(fast && fast->next)
    {
        slow=slow->next;
        fast=fast->next->next;

        if(slow==fast)
            return true;
    }

    return false;
}

int main()
{
    Node *head=new Node{10,NULL};
```


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```
head->next=new Node{20,NULL};
head->next->next=new Node{30,NULL};
head->next->next->next=new Node{40,NULL};

head->next->next->next->next=head->next;

if(detectLoop(head))
    cout<<"Loop Found";
else
    cout<<"No Loop";
}
```

Time Complexity: $O(n)$

Q97. Remove Loop from Linked List

C++ Solution

```
#include<iostream>
using namespace std;

struct Node
{
    int data;
    Node *next;
};

void removeLoop(Node *head)
{
    Node *slow=head;
    Node *fast=head;

    while(fast && fast->next)
    {
        slow=slow->next;
        fast=fast->next->next;

        if(slow==fast)
            break;
    }

    if(slow!=fast)
        return;

    slow=head;

    while(slow->next!=fast->next)
    {
        slow=slow->next;
```

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```
        fast=fast->next;
    }

    fast->next=NULL;
}

int main()
{
    cout<<"Loop Removed";
}
```

Time Complexity: $O(n)$

Q98. Merge Two Sorted Linked Lists

Example

Input

1→3→5
2→4→6

Output

1→2→3→4→5→6

C++ Solution

```
#include<iostream>
using namespace std;

struct Node
{
    int data;
    Node *next;
};

Node* merge(Node *a,Node *b)
{
    if(a==NULL)
        return b;

    if(b==NULL)
        return a;

    if(a->data<b->data)
    {
        a->next=merge(a->next,b);
        return a;
    }
```

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```
b->next=merge(a,b->next);
return b;
}

int main()
{
    cout<<"Merged Successfully";
}
```

Time Complexity: $O(n)$

Q99. Remove Duplicates from Sorted Linked List

Example

Input

10→10→20→20→30

Output

10→20→30

C++ Solution

```
#include<iostream>
using namespace std;

struct Node
{
    int data;
    Node *next;
};

void removeDuplicate(Node *head)
{
    while(head && head->next)
    {
        if(head->data==head->next->data)
        {
            Node *temp=head->next;

            head->next=temp->next;

            delete temp;
        }
        else
            head=head->next;
    }
}
```

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```
}

int main()
{
    cout<<"Duplicates Removed";
}
```

Time Complexity: $O(n)$

Q100. Check Palindrome Linked List

C++ Solution

```
#include<iostream>
#include<vector>
using namespace std;

struct Node
{
    int data;
    Node *next;
};

bool palindrome(Node *head)
{
    vector<int> v;

    while(head)
    {
        v.push_back(head->data);

        head=head->next;
    }

    int i=0,j=v.size()-1;

    while(i<j)
    {
        if(v[i]!=v[j])
            return false;

        i++;
        j--;
    }

    return true;
}

int main()
```

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```
{  
    cout<<"Palindrome";  
}
```

Time Complexity: $O(n)$

Q101. Intersection of Two Linked Lists

C++ Solution

```
#include<iostream>  
using namespace std;
```

```
struct Node  
{  
    int data;  
    Node *next;  
};
```

```
Node* intersection(Node *a,Node *b)  
{  
    Node *p=a;  
    Node *q=b;  
  
    while(p!=q)  
    {  
        p=(p==NULL)?b:p->next;  
        q=(q==NULL)?a:q->next;  
    }  
  
    return p;  
}
```

```
int main()  
{  
    cout<<"Intersection Found";  
}
```

Time Complexity: $O(n)$

Q102. Circular Linked List Traversal

C++ Solution

```
#include<iostream>  
using namespace std;
```

```
struct Node
```

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```
{
    int data;
    Node *next;
};

int main()
{
    Node *head=new Node{10,NULL};
    Node *second=new Node{20,NULL};
    Node *third=new Node{30,NULL};

    head->next=second;
    second->next=third;
    third->next=head;

    Node *temp=head;

    do
    {
        cout<<temp->data<<" ";

        temp=temp->next;

    }while(temp!=head);
}
```

Time Complexity: $O(n)$

Q103. Doubly Linked List Traversal

C++ Solution

```
#include<iostream>
using namespace std;

struct Node
{
    int data;
    Node *prev;
    Node *next;
};

int main()
{
    Node *first=new Node{10,NULL,NULL};
    Node *second=new Node{20,first,NULL};
    first->next=second;

    Node *temp=first;
```

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```
while(temp)
{
    cout<<temp->data<<" ";

    temp=temp->next;
}
}
```

Time Complexity: $O(n)$

Q104. Reverse Linked List in Groups of K

C++ Solution

```
#include<iostream>
using namespace std;

struct Node
{
    int data;
    Node *next;
};

Node* reverse(Node *head,int k)
{
    Node *curr=head;
    Node *prev=NULL;
    Node *next=NULL;

    int count=0;

    while(curr && count<k)
    {
        next=curr->next;
        curr->next=prev;
        prev=curr;
        curr=next;
        count++;
    }

    if(next)
        head->next=reverse(next,k);

    return prev;
}

int main()
{
```

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```
    cout<<"Linked List Reversed";  
}
```

Time Complexity: $O(n)$

Q105. Delete Node Without Head Pointer

Example

Input

10 → 20 → 30 → 40
Delete Node = 30

Output

10 → 20 → 40

C++ Solution

```
#include<iostream>  
using namespace std;  
  
struct Node  
{  
    int data;  
    Node *next;  
};  
  
void deleteNode(Node *node)  
{  
    Node *temp=node->next;  
  
    node->data=temp->data;  
  
    node->next=temp->next;  
  
    delete temp;  
}  
  
int main()  
{  
    cout<<"Node Deleted";  
}
```

Time Complexity: $O(1)$

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Linked List Section Completed (20/20)

Q106. Implement Stack Using Array

Problem: Implement push(), pop(), peek(), and display() using an array.

C++ Solution

```
#include<iostream>
using namespace std;

#define MAX 100

class Stack
{
    int arr[MAX];
    int top;

public:

    Stack()
    {
        top=-1;
    }

    void push(int x)
    {
        if(top==MAX-1)
        {
            cout<<"Overflow";
            return;
        }

        arr[++top]=x;
    }

    void pop()
    {
        if(top==-1)
        {
            cout<<"Underflow";
            return;
        }

        top--;
    }

    void peek()
    {
        if(top!=-1)
```

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```
        cout<<arr[top];
    }

    void display()
    {
        for(int i=top;i>=0;i--)
            cout<<arr[i]<<" ";
    }
};

int main()
{
    Stack s;

    s.push(10);
    s.push(20);
    s.push(30);

    s.display();
}
```

Time Complexity

Push $\rightarrow O(1)$

Pop $\rightarrow O(1)$

Peek $\rightarrow O(1)$

Q107. Implement Stack Using Linked List

```
#include<iostream>
using namespace std;
```

```
struct Node
{
    int data;
    Node *next;
};
```

```
class Stack
{
    Node *top;
```

public:

```
    Stack()
    {
```

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```
    top=NULL;
}

void push(int x)
{
    Node *temp=new Node;

    temp->data=x;
    temp->next=top;

    top=temp;
}

void pop()
{
    if(top==NULL)
        return;

    Node *temp=top;

    top=top->next;

    delete temp;
}

void display()
{
    Node *temp=top;

    while(temp)
    {
        cout<<temp->data<<" ";

        temp=temp->next;
    }
}

int main()
{
    Stack s;

    s.push(5);
    s.push(10);
    s.push(15);

    s.display();
}
```

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Q108. Implement Queue Using Array

```
#include<iostream>
using namespace std;

#define SIZE 100

class Queue
{
    int arr[SIZE];
    int front,rear;

public:

    Queue()
    {
        front=0;
        rear=-1;
    }

    void enqueue(int x)
    {
        arr[++rear]=x;
    }

    void dequeue()
    {
        if(front<=rear)
            front++;
    }

    void display()
    {
        for(int i=front;i<=rear;i++)
            cout<<arr[i]<<" ";
    }
};

int main()
{
    Queue q;

    q.enqueue(10);
    q.enqueue(20);
    q.enqueue(30);

    q.display();
}
```

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Q109. Circular Queue

```
#include<iostream>
using namespace std;

#define SIZE 5

class CircularQueue
{
    int arr[SIZE];
    int front,rear;

public:

    CircularQueue()
    {
        front=-1;
        rear=-1;
    }

    void enqueue(int x)
    {
        if((rear+1)%SIZE==front)
        {
            cout<<"Full";
            return;
        }

        if(front==-1)
            front=0;

        rear=(rear+1)%SIZE;

        arr[rear]=x;
    }

    void display()
    {
        int i=front;

        while(true)
        {
            cout<<arr[i]<<" ";

            if(i==rear)
                break;

            i=(i+1)%SIZE;
        }
    }
}
```

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```
};

int main()
{
    CircularQueue q;

    q.enqueue(10);
    q.enqueue(20);
    q.enqueue(30);

    q.display();
}
```

Q110. Implement Queue Using Two Stacks

```
#include<iostream>
#include<stack>
using namespace std;

stack<int> s1,s2;

void enqueue(int x)
{
    s1.push(x);
}

void dequeue()
{
    if(s2.empty())
    {
        while(!s1.empty())
        {
            s2.push(s1.top());
            s1.pop();
        }
    }

    if(!s2.empty())
        s2.pop();
}

int main()
{
    enqueue(10);
    enqueue(20);
    enqueue(30);

    dequeue();
}
```

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```
    cout<<s2.top();  
}
```

Q111. Implement Stack Using Two Queues

```
#include<iostream>  
#include<queue>  
using namespace std;  
  
queue<int> q1,q2;  
  
void push(int x)  
{  
    q2.push(x);  
  
    while(!q1.empty())  
    {  
        q2.push(q1.front());  
        q1.pop();  
    }  
  
    swap(q1,q2);  
}  
  
void pop()  
{  
    if(!q1.empty())  
        q1.pop();  
}  
  
int main()  
{  
    push(10);  
    push(20);  
    push(30);  
  
    pop();  
  
    cout<<q1.front();  
}
```

Q112. Parentheses Matching

```
#include<iostream>  
#include<stack>  
using namespace std;  
  
int main()
```

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```
{
    string s;

    cin>>s;

    stack<char> st;

    for(char c:s)
    {
        if(c=='('||c=='{'||c=='[')
            st.push(c);

        else
        {
            if(st.empty())
            {
                cout<<"Invalid";
                return 0;
            }

            if((c==')'&&st.top()!='(')||
               (c=='}'&&st.top()!='{'||
               (c==']'&&st.top()!='['))
            {
                cout<<"Invalid";
                return 0;
            }

            st.pop();
        }
    }

    cout<<(st.empty()?"Valid":"Invalid");
}
```

Q113. Infix to Postfix

```
#include<iostream>
#include<stack>
using namespace std;

int priority(char c)
{
    if(c=='+'||c=='-')
        return 1;

    if(c=='*'||c=='/')
        return 2;
```


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```
    if(c=='^')
        return 3;

    return 0;
}

int main()
{
    string exp;

    cin>>exp;

    stack<char> st;

    string ans="";

    for(char c:exp)
    {
        if(isalnum(c))
            ans+=c;

        else if(c=='(')
            st.push(c);

        else if(c==')')
        {
            while(st.top()!='(')
            {
                ans+=st.top();
                st.pop();
            }

            st.pop();
        }

        else
        {
            while(!st.empty() && priority(st.top())>=priority(c))
            {
                ans+=st.top();
                st.pop();
            }

            st.push(c);
        }
    }

    while(!st.empty())
    {
        ans+=st.top();
```

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```
        st.pop();
    }

    cout<<ans;
}
```

Q114. Postfix Expression Evaluation

```
#include<iostream>
#include<stack>
using namespace std;

int main()
{
    string exp;

    cin>>exp;

    stack<int> st;

    for(char c:exp)
    {
        if(isdigit(c))
            st.push(c-'0');

        else
        {
            int b=st.top();
            st.pop();

            int a=st.top();
            st.pop();

            switch(c)
            {
                case '+': st.push(a+b); break;
                case '-': st.push(a-b); break;
                case '*': st.push(a*b); break;
                case '/': st.push(a/b); break;
            }
        }
    }

    cout<<st.top();
}
```

Q115. Reverse a Stack Using Recursion

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```
#include<iostream>
#include<stack>
using namespace std;

void insertBottom(stack<int> &st,int x)
{
    if(st.empty())
    {
        st.push(x);
        return;
    }

    int t=st.top();

    st.pop();

    insertBottom(st,x);

    st.push(t);
}

void reverse(stack<int> &st)
{
    if(st.empty())
        return;

    int x=st.top();

    st.pop();

    reverse(st);

    insertBottom(st,x);
}

int main()
{
    stack<int> st;

    st.push(1);
    st.push(2);
    st.push(3);

    reverse(st);

    while(!st.empty())
    {
        cout<<st.top()<<" ";
        st.pop();
    }
}
```

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```
}  
}
```

Q116. Next Greater Element

Problem: For every element in the array, find the next greater element on its right.

Example

Input

4
4 5 2 25

Output

5 25 25 -1

C++ Solution

```
#include<iostream>  
#include<stack>  
using namespace std;  
  
int main()  
{  
    int n;  
    cin>>n;  
  
    int arr[n],ans[n];  
  
    for(int i=0;i<n;i++)  
        cin>>arr[i];  
  
    stack<int> st;  
  
    for(int i=n-1;i>=0;i--)  
    {  
        while(!st.empty() && st.top()<=arr[i])  
            st.pop();  
  
        ans[i]=st.empty()? -1:st.top();  
  
        st.push(arr[i]);  
    }  
  
    for(int i=0;i<n;i++)  
        cout<<ans[i]<<" ";  
}
```

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Time Complexity: $O(n)$

Q117. Stock Span Problem

Example

Input

7
100 80 60 70 60 75 85

Output

1 1 1 2 1 4 6

C++ Solution

```
#include<iostream>
#include<stack>
using namespace std;

int main()
{
    int n;
    cin>>n;

    int price[n];

    for(int i=0;i<n;i++)
        cin>>price[i];

    stack<int> st;

    for(int i=0;i<n;i++)
    {
        while(!st.empty() && price[st.top()]<=price[i])
            st.pop();

        if(st.empty())
            cout<<i+1<<" ";
        else
            cout<<i-st.top()<<" ";

        st.push(i);
    }
}
```

Time Complexity: $O(n)$

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Q118. Largest Rectangle in Histogram

Example

Input

6
2 1 5 6 2 3

Output

10

C++ Solution

```
#include<iostream>
#include<stack>
using namespace std;

int main()
{
    int n;
    cin>>n;

    int h[n];

    for(int i=0;i<n;i++)
        cin>>h[i];

    stack<int> st;

    int ans=0;

    for(int i=0;i<=n;i++)
    {
        while(!st.empty() && (i==n || h[st.top()]>=h[i]))
        {
            int height=h[st.top()];
            st.pop();

            int width=st.empty()?i:st.top()-1;

            ans=max(ans,height*width);
        }

        st.push(i);
    }
}
```

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```
    cout<<ans;  
}
```

Time Complexity: $O(n)$

Q119. Sliding Window Maximum

Example

Input

```
8  
1 3 -1 -3 5 3 6 7  
3
```

Output

```
3 3 5 5 6 7
```

C++ Solution

```
#include<iostream>  
#include<deque>  
using namespace std;  
  
int main()  
{  
    int n,k;  
  
    cin>>n;  
  
    int arr[n];  
  
    for(int i=0;i<n;i++)  
        cin>>arr[i];  
  
    cin>>k;  
  
    deque<int> dq;  
  
    for(int i=0;i<n;i++)  
    {  
        while(!dq.empty() && dq.front()<=i-k)  
            dq.pop_front();  
  
        while(!dq.empty() && arr[dq.back()]<=arr[i])  
            dq.pop_back();  
  
        dq.push_back(i);  
  
        if(dq.size()==k)  
            cout<<arr[dq.front()]<<" ";  
    }  
}
```

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```
        dq.push_back(i);

        if(i>=k-1)
            cout<<arr[dq.front()]<<" ";
    }
}
```

Time Complexity: $O(n)$

Q120. Min Stack

Problem: Design a stack that supports push(), pop(), and getMin() in $O(1)$.

C++ Solution

```
#include<iostream>
#include<stack>
using namespace std;

stack<int> st,minSt;

void push(int x)
{
    st.push(x);

    if(minSt.empty() || x<=minSt.top())
        minSt.push(x);
}

void pop()
{
    if(st.top()==minSt.top())
        minSt.pop();

    st.pop();
}

int getMin()
{
    return minSt.top();
}

int main()
{
    push(5);
    push(2);
    push(8);
```


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```
    cout<<getMin();  
}
```

Time Complexity: $O(1)$

Q121. Implement Deque

C++ Solution

```
#include<iostream>  
#include<deque>  
using namespace std;  
  
int main()  
{  
    deque<int> dq;  
  
    dq.push_front(10);  
    dq.push_back(20);  
    dq.push_front(5);  
  
    while(!dq.empty())  
    {  
        cout<<dq.front()<<" ";  
  
        dq.pop_front();  
    }  
}
```

Time Complexity: $O(1)$

Q122. Reverse a Queue

C++ Solution

```
#include<iostream>  
#include<queue>  
#include<stack>  
using namespace std;  
  
int main()  
{  
    queue<int> q;  
  
    q.push(10);  
    q.push(20);
```

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```
q.push(30);

stack<int> st;

while(!q.empty())
{
    st.push(q.front());
    q.pop();
}

while(!st.empty())
{
    q.push(st.top());
    st.pop();
}

while(!q.empty())
{
    cout<<q.front()<<" ";
    q.pop();
}
}
```

Time Complexity: $O(n)$

Q123. First Non-Repeating Character in a Stream

Example

Input

aabc

Output

a -1 b b

C++ Solution

```
#include<iostream>
#include<queue>
using namespace std;

int main()
{
    string s;
    cin>>s;
```

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```
int freq[26]={0};

queue<char> q;

for(char ch:s)
{
    freq[ch-'a']++;

    q.push(ch);

    while(!q.empty() && freq[q.front()-'a']>1)
        q.pop();

    if(q.empty())
        cout<<"-1 ";
    else
        cout<<q.front()<<" ";
}
}
```

Time Complexity: $O(n)$

Q124. LRU Cache (Basic Implementation)

C++ Solution

```
#include<iostream>
#include<list>
using namespace std;

int main()
{
    list<int> cache;

    cache.push_front(10);
    cache.push_front(20);
    cache.push_front(30);

    for(int x:cache)
        cout<<x<<" ";
}
```

Time Complexity

Access: $O(1)$

Insertion: $O(1)$

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Deletion: $O(1)$

Q125. Tower of Hanoi

Example

Input

3

Output

Move Disk 1 from A to C
Move Disk 2 from A to B
Move Disk 1 from C to B
Move Disk 3 from A to C
Move Disk 1 from B to A
Move Disk 2 from B to C
Move Disk 1 from A to C

C++ Solution

```
#include<iostream>
using namespace std;

void tower(int n,char source,char helper,char destination)
{
    if(n==1)
    {
        cout<<"Move Disk 1 from "<<source<<" to "<<destination<<endl;
        return;
    }

    tower(n-1,source,destination,helper);

    cout<<"Move Disk "<<n<<" from "<<source<<" to "<<destination<<endl;

    tower(n-1,helper,source,destination);
}

int main()
{
    int n;
    cin>>n;

    tower(n,'A','B','C');
}
```

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Time Complexity: $O(2^n)$

Trees Coding Questions (Q126–Q145)

Q126. Inorder Traversal

Problem: Print the nodes of a binary tree in Inorder (Left $\rightarrow$ Root $\rightarrow$ Right).

Example

Input

```
    1
   / \
  2   3
 / \
4   5
```

Output

4 2 5 1 3

C++ Solution

```
#include<iostream>
using namespace std;

struct Node
{
    int data;
    Node *left,*right;

    Node(int x)
    {
        data=x;
        left=right=NULL;
    }
};

void inorder(Node *root)
{
    if(root==NULL)
        return;

    inorder(root->left);
    cout<<root->data<<" ";
    inorder(root->right);
}

int main()
```

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```
{
    Node *root=new Node(1);
    root->left=new Node(2);
    root->right=new Node(3);
    root->left->left=new Node(4);
    root->left->right=new Node(5);

    inorder(root);
}
```

Q127. Preorder Traversal

Output

```
1 2 4 5 3
void preorder(Node *root)
{
    if(root==NULL)
        return;

    cout<<root->data<<" ";
    preorder(root->left);
    preorder(root->right);
}
```

Q128. Postorder Traversal

Output

```
4 5 2 3 1
void postorder(Node *root)
{
    if(root==NULL)
        return;

    postorder(root->left);
    postorder(root->right);
    cout<<root->data<<" ";
}
```

Q129. Level Order Traversal

```
#include<iostream>
#include<queue>
using namespace std;

void levelOrder(Node *root)
{
    queue<Node*> q;
```

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```
q.push(root);

while(!q.empty())
{
    Node *temp=q.front();
    q.pop();

    cout<<temp->data<<" ";

    if(temp->left)
        q.push(temp->left);

    if(temp->right)
        q.push(temp->right);
}
}
```

Q130. Height of Binary Tree

```
int height(Node *root)
{
    if(root==NULL)
        return 0;

    return 1+max(height(root->left),height(root->right));
}
```

Q131. Count Total Nodes

```
int count(Node *root)
{
    if(root==NULL)
        return 0;

    return 1+count(root->left)+count(root->right);
}
```

Q132. Count Leaf Nodes

```
int leaf(Node *root)
{
    if(root==NULL)
        return 0;

    if(root->left==NULL && root->right==NULL)
        return 1;

    return leaf(root->left)+leaf(root->right);
}
```

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Q133. Search Element in Binary Tree

```
bool search(Node *root,int key)
{
    if(root==NULL)
        return false;

    if(root->data==key)
        return true;

    return search(root->left,key)||search(root->right,key);
}
```

Q134. Mirror Binary Tree

```
void mirror(Node *root)
{
    if(root==NULL)
        return;

    swap(root->left,root->right);

    mirror(root->left);
    mirror(root->right);
}
```

Q135. Check Identical Trees

```
bool identical(Node *a,Node *b)
{
    if(a==NULL && b==NULL)
        return true;

    if(a==NULL || b==NULL)
        return false;

    return a->data==b->data &&
           identical(a->left,b->left) &&
           identical(a->right,b->right);
}
```

Q136. Lowest Common Ancestor (Binary Tree)

```
Node* lca(Node *root,int a,int b)
{
    if(root==NULL)
        return NULL;

    if(root->data==a||root->data==b)
        return root;

    Node *left=lca(root->left,a,b);
```


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```
Node *right=lca(root->right,a,b);

if(left && right)
    return root;

return left?left:right;
}
```

Q137. Diameter of Binary Tree

```
int height(Node *root)
{
    if(root==NULL)
        return 0;

    return 1+max(height(root->left),height(root->right));
}

int diameter(Node *root)
{
    if(root==NULL)
        return 0;

    int op1=height(root->left)+height(root->right)+1;
    int op2=diameter(root->left);
    int op3=diameter(root->right);

    return max(op1,max(op2,op3));
}
```

Q138. Left View of Binary Tree

```
#include<queue>

void leftView(Node *root)
{
    queue<Node*> q;

    q.push(root);

    while(!q.empty())
    {
        int size=q.size();

        for(int i=0;i<size;i++)
        {
            Node *temp=q.front();

            q.pop();

            if(i==0)
```

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```
        cout<<temp->data<<" ";

        if(temp->left)
            q.push(temp->left);

        if(temp->right)
            q.push(temp->right);
    }
}
}
```

Q139. Right View of Binary Tree

```
#include<queue>
```

```
void rightView(Node *root)
{
    queue<Node*> q;

    q.push(root);

    while(!q.empty())
    {
        int size=q.size();

        for(int i=0;i<size;i++)
        {
            Node *temp=q.front();

            q.pop();

            if(i==size-1)
                cout<<temp->data<<" ";

            if(temp->left)
                q.push(temp->left);

            if(temp->right)
                q.push(temp->right);
        }
    }
}
```

Q140. Check Balanced Binary Tree

```
int height(Node *root)
{
    if(root==NULL)
        return 0;

    return 1+max(height(root->left),height(root->right));
}
```

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```
}

bool balanced(Node *root)
{
    if(root==NULL)
        return true;

    int lh=height(root->left);
    int rh=height(root->right);

    if(abs(lh-rh)>1)
        return false;

    return balanced(root->left)&&balanced(root->right);
}
```

Q141. BST Insertion

```
Node* insert(Node *root,int key)
{
    if(root==NULL)
        return new Node(key);

    if(key<root->data)
        root->left=insert(root->left,key);

    else
        root->right=insert(root->right,key);

    return root;
}
```

Q142. BST Search

```
bool search(Node *root,int key)
{
    if(root==NULL)
        return false;

    if(root->data==key)
        return true;

    if(key<root->data)
        return search(root->left,key);

    return search(root->right,key);
}
```

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Q143. BST Deletion

```
Node* minValue(Node *root)
{
    while(root->left)
        root=root->left;

    return root;
}

Node* deleteNode(Node *root,int key)
{
    if(root==NULL)
        return NULL;

    if(key<root->data)
        root->left=deleteNode(root->left,key);

    else if(key>root->data)
        root->right=deleteNode(root->right,key);

    else
    {
        if(root->left==NULL)
            return root->right;

        if(root->right==NULL)
            return root->left;

        Node *temp=minValue(root->right);

        root->data=temp->data;

        root->right=deleteNode(root->right,temp->data);
    }

    return root;
}
```

Q144. Validate Binary Search Tree

```
bool valid(Node *root,long long mn,long long mx)
{
    if(root==NULL)
        return true;

    if(root->data<=mn||root->data>=mx)
        return false;

    return valid(root->left,mn,root->data) &&
```

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```
        valid(root->right,root->data,mx);
    }
```

Q145. Kth Smallest Element in BST

```
void inorder(Node *root,int &k,int &ans)
{
    if(root==NULL)
        return;

    inorder(root->left,k,ans);

    k--;

    if(k==0)
    {
        ans=root->data;
        return;
    }

    inorder(root->right,k,ans);
}
```

Graph Coding Questions (Q146–Q160)

Q146. Breadth First Search (BFS)

Problem: Traverse a graph using Breadth First Search.

Example

Input

```
5 4
0 1
0 2
1 3
2 4
```

Output

```
0 1 2 3 4
```

C++ Solution

```
#include<iostream>
#include<vector>
#include<queue>
```

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```
using namespace std;

int main()
{
    int V,E;
    cin>>V>>E;

    vector<int> adj[V];

    for(int i=0;i<E;i++)
    {
        int u,v;
        cin>>u>>v;

        adj[u].push_back(v);
        adj[v].push_back(u);
    }

    vector<bool> visited(V,false);

    queue<int> q;

    q.push(0);
    visited[0]=true;

    while(!q.empty())
    {
        int node=q.front();
        q.pop();

        cout<<node<<" ";

        for(int x:adj[node])
        {
            if(!visited[x])
            {
                visited[x]=true;
                q.push(x);
            }
        }
    }
}
```

Time Complexity: **$O(V+E)$**

Q147. Depth First Search (DFS)

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```
#include<iostream>
#include<vector>
using namespace std;

void dfs(int node,vector<int> adj[],vector<bool>& vis)
{
    vis[node]=true;

    cout<<node<<" ";

    for(int x:adj[node])
        if(!vis[x])
            dfs(x,adj,vis);
}

int main()
{
    int V=5;

    vector<int> adj[V];

    adj[0]={1,2};
    adj[1]={3};
    adj[2]={4};

    vector<bool> vis(V,false);

    dfs(0,adj,vis);
}
```

Time Complexity: **$O(V+E)$**

Q148. Detect Cycle in Undirected Graph (DFS)

```
#include<iostream>
#include<vector>
using namespace std;

bool cycle(int node,int parent,vector<int> adj[],vector<bool>& vis)
{
    vis[node]=true;

    for(int x:adj[node])
    {
        if(!vis[x])
        {
            if(cycle(x,node,adj,vis))
                return true;
        }
    }
}
```

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```
    }

    else if(x!=parent)
        return true;
    }

    return false;
}
```

Time Complexity: **O(V+E)**

Q149. Detect Cycle in Directed Graph

```
#include<iostream>
#include<vector>
using namespace std;

bool dfs(int node,vector<int> adj[],
vector<int>& vis,vector<int>& path)
{
    vis[node]=1;
    path[node]=1;

    for(int x:adj[node])
    {
        if(!vis[x])
        {
            if(dfs(x,adj,vis,path))
                return true;
        }

        else if(path[x])
            return true;
    }

    path[node]=0;

    return false;
}
```

Time Complexity: **O(V+E)**

Q150. Topological Sort (DFS)

```
#include<iostream>
#include<vector>
```


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```
#include<stack>
using namespace std;

void topo(int node,
vector<int> adj[],
vector<bool>& vis,
stack<int>& st)
{
    vis[node]=true;

    for(int x:adj[node])

        if(!vis[x])

            topo(x,adj,vis,st);

    st.push(node);
}
```

Time Complexity: **O(V+E)**

Q151. Topological Sort (Kahn's Algorithm)

```
#include<iostream>
#include<vector>
#include<queue>
using namespace std;

int main()
{
    int V=6;

    vector<int> adj[V];

    vector<int> indegree(V,0);

    queue<int> q;

    for(int i=0;i<V;i++)
        if(indegree[i]==0)
            q.push(i);

    while(!q.empty())
    {
        int node=q.front();

        q.pop();
```

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```
    cout<<node<<" ";

    for(int x:adj[node])
    {
        indegree[x]--;

        if(indegree[x]==0)
            q.push(x);
    }
}
```

Q152. Dijkstra's Algorithm

```
#include<iostream>
#include<vector>
#include<queue>
using namespace std;

typedef pair<int,int> pii;

int main()
{
    int V=5;

    vector<pii> adj[V];

    priority_queue<pii,
    vector<pii>,
    greater<pii>> pq;

    vector<int> dist(V,1e9);

    dist[0]=0;

    pq.push({0,0});

    while(!pq.empty())
    {
        auto it=pq.top();

        pq.pop();

        int d=it.first;
        int node=it.second;

        for(auto x:adj[node])
        {
            if(d+x.second<dist[x.first])
```

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```
        {
            dist[x.first]=d+x.second;

            pq.push({dist[x.first],x.first});
        }
    }
}
```

Time Complexity: **$O(E \log V)$**

Q153. Prim's Algorithm

```
#include<iostream>
#include<vector>
#include<queue>
using namespace std;

typedef pair<int,int> pii;

int main()
{
    int V=5;

    vector<pii> adj[V];

    priority_queue<pii,
vector<pii>,
greater<pii>> pq;

    vector<int> vis(V,false);

    pq.push({0,0});

    int cost=0;

    while(!pq.empty())
    {
        auto p=pq.top();

        pq.pop();

        int wt=p.first;
        int node=p.second;

        if(vis[node])
            continue;
```

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```
vis[node]=true;

cost+=wt;

for(auto x:adj[node])

    if(!vis[x.first])

        pq.push({x.second,x.first});
}

cout<<cost;
}
```

Q154. Kruskal's Algorithm

```
#include<iostream>
#include<vector>
#include<algorithm>
using namespace std;

struct Edge
{
    int u,v,w;
};

bool cmp(Edge a,Edge b)
{
    return a.w<b.w;
}

int main()
{
    vector<Edge> edge;

    sort(edge.begin(),edge.end(),cmp);
}
```

Time Complexity: **$O(E \log E)$**

Q155. Disjoint Set Union (Union Find)

```
#include<iostream>
using namespace std;

int parent[100];
```

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```
int find(int x)
{
    if(parent[x]==x)
        return x;

    return parent[x]=find(parent[x]);
}

void unite(int a,int b)
{
    a=find(a);

    b=find(b);

    parent[a]=b;
}
```

Time Complexity: Nearly **O(1)**

Q156. Number of Connected Components

```
#include<iostream>
#include<vector>
using namespace std;

void dfs(int node,
vector<int> adj[],
vector<bool>& vis)
{
    vis[node]=true;

    for(int x:adj[node])

        if(!vis[x])

            dfs(x,adj,vis);
}
```

Q157. Bipartite Graph Check

```
#include<iostream>
#include<vector>
#include<queue>
using namespace std;

bool bipartite(vector<int> adj[],int V)
{
}
```

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```
vector<int> color(V,-1);

queue<int> q;

q.push(0);

color[0]=0;

while(!q.empty())
{
    int node=q.front();

    q.pop();

    for(int x:adj[node])
    {
        if(color[x]==-1)
        {
            color[x]=1-color[node];

            q.push(x);
        }

        else if(color[x]==color[node])

            return false;
    }
}

return true;
}
```

Q158. Shortest Path in Unweighted Graph

```
#include<iostream>
#include<vector>
#include<queue>
using namespace std;

int main()
{
    int V=6;

    vector<int> adj[V];

    vector<int> dist(V,-1);

    queue<int> q;
```

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```
q.push(0);

dist[0]=0;

while(!q.empty())
{
    int node=q.front();

    q.pop();

    for(int x:adj[node])
    {
        if(dist[x]==-1)
        {
            dist[x]=dist[node]+1;

            q.push(x);
        }
    }
}
```

Q159. Flood Fill Algorithm

```
#include<iostream>
#include<vector>
using namespace std;

void dfs(vector<vector<int>>& image,
int r,int c,
int oldColor,
int newColor)
{
    if(r<0||c<0||
        r>=image.size()||
        c>=image[0].size())
        return;

    if(image[r][c]!=oldColor)
        return;

    image[r][c]=newColor;

    dfs(image,r+1,c,oldColor,newColor);
    dfs(image,r-1,c,oldColor,newColor);
    dfs(image,r,c+1,oldColor,newColor);
    dfs(image,r,c-1,oldColor,newColor);
}
```

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Q160. Number of Islands

```
#include<iostream>
#include<vector>
using namespace std;

void dfs(vector<vector<char>>& grid,
int i,int j)
{
    if(i<0||j<0||
        i>=grid.size()||
        j>=grid[0].size()||
        grid[i][j]=='0')
        return;

    grid[i][j]='0';

    dfs(grid,i+1,j);
    dfs(grid,i-1,j);
    dfs(grid,i,j+1);
    dfs(grid,i,j-1);
}
```

Time Complexity: **$O(m \times n)$**

Dynamic Programming Coding Questions (Q161–Q180)

Q161. Fibonacci Number (Memoization)

Problem: Find the nth Fibonacci number using Dynamic Programming.

Example

Input

6

Output

8

C++ Solution

```
#include<iostream>
#include<vector>
using namespace std;

int fib(int n, vector<int> &dp)
```


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```
{
    if(n<=1)
        return n;

    if(dp[n]!=-1)
        return dp[n];

    return dp[n]=fib(n-1,dp)+fib(n-2,dp);
}

int main()
{
    int n;
    cin>>n;

    vector<int> dp(n+1,-1);

    cout<<fib(n,dp);
}
```

Time Complexity: **O(n)**

Q162. Fibonacci (Tabulation)

```
#include<iostream>
#include<vector>
using namespace std;

int main()
{
    int n;
    cin>>n;

    vector<int> dp(n+1);

    dp[0]=0;
    dp[1]=1;

    for(int i=2;i<=n;i++)
        dp[i]=dp[i-1]+dp[i-2];

    cout<<dp[n];
}
```

Time Complexity: **O(n)**

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Q163. Climbing Stairs

Example

Input

5

Output

8

```
#include<iostream>
#include<vector>
using namespace std;
```

```
int main()
{
    int n;
    cin>>n;

    vector<int> dp(n+1);

    dp[0]=1;
    dp[1]=1;

    for(int i=2;i<=n;i++)
        dp[i]=dp[i-1]+dp[i-2];

    cout<<dp[n];
}
```

Time Complexity: **O(n)**

Q164. 0/1 Knapsack

```
#include<iostream>
using namespace std;
```

```
int main()
{
    int wt[]={1,3,4,5};
    int val[]={1,4,5,7};

    int W=7;
    int n=4;

    int dp[5][8]={0};
```

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```
for(int i=1;i<=n;i++)
{
    for(int j=1;j<=W;j++)
    {
        if(wt[i-1]<=j)
            dp[i][j]=max(val[i-1]+dp[i-1][j-wt[i-1]],dp[i-1][j]);
        else
            dp[i][j]=dp[i-1][j];
    }
}

cout<<dp[n][W];
}
```

Time Complexity: **$O(n \times W)$**

Q165. Subset Sum Problem

```
#include<iostream>
using namespace std;

int main()
{
    int arr[]={3,34,4,12,5,2};

    int sum=9;
    int n=6;

    bool dp[7][10]={false};

    for(int i=0;i<=n;i++)
        dp[i][0]=true;

    for(int i=1;i<=n;i++)
    {
        for(int j=1;j<=sum;j++)
        {
            if(arr[i-1]<=j)
                dp[i][j]=dp[i-1][j]||dp[i-1][j-arr[i-1]];
            else
                dp[i][j]=dp[i-1][j];
        }
    }

    cout<<dp[n][sum];
}
```

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Q166. Equal Partition

```
#include<iostream>
#include<vector>
using namespace std;

bool subset(vector<int>& arr,int sum)
{
    int n=arr.size();

    vector<vector<bool>> dp(n+1,vector<bool>(sum+1,false));

    for(int i=0;i<=n;i++)
        dp[i][0]=true;

    for(int i=1;i<=n;i++)
    {
        for(int j=1;j<=sum;j++)
        {
            if(arr[i-1]<=j)
                dp[i][j]=dp[i-1][j]||dp[i-1][j-arr[i-1]];
            else
                dp[i][j]=dp[i-1][j];
        }
    }

    return dp[n][sum];
}

int main()
{
    vector<int> arr={1,5,11,5};

    int sum=22;

    cout<<subset(arr,sum/2);
}
```

Q167. Coin Change (Minimum Coins)

```
#include<iostream>
#include<vector>
using namespace std;

int main()
{
    vector<int> coin={1,2,5};

    int amount=11;
```

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```
vector<int> dp(amount+1,1e9);

dp[0]=0;

for(int i=1;i<=amount;i++)
{
    for(int c:coin)
    {
        if(c<=i)
            dp[i]=min(dp[i],1+dp[i-c]);
    }
}

cout<<dp[amount];
}
```

Q168. Coin Change (Number of Ways)

```
#include<iostream>
using namespace std;

int main()
{
    int coin[]={1,2,3};

    int sum=4;

    int dp[5]={0};

    dp[0]=1;

    for(int i=0;i<3;i++)
    {
        for(int j=coin[i];j<=sum;j++)
            dp[j]+=dp[j-coin[i]];
    }

    cout<<dp[sum];
}
```

Q169. Longest Common Subsequence (LCS)

```
#include<iostream>
using namespace std;

int main()
{
```

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```
string a="ABCB DAB";
string b="BDCAB";

int dp[8][6]={0};

for(int i=1;i<=a.length();i++)
{
    for(int j=1;j<=b.length();j++)
    {
        if(a[i-1]==b[j-1])
            dp[i][j]=1+dp[i-1][j-1];
        else
            dp[i][j]=max(dp[i-1][j],dp[i][j-1]);
    }
}

cout<<dp[a.length()][b.length()];
}
```

Q170. Longest Common Substring

```
#include<iostream>
using namespace std;

int main()
{
    string a="abcdxyz";
    string b="xyzabcd";

    int dp[20][20]={0};

    int ans=0;

    for(int i=1;i<=a.length();i++)
    {
        for(int j=1;j<=b.length();j++)
        {
            if(a[i-1]==b[j-1])
            {
                dp[i][j]=1+dp[i-1][j-1];

                ans=max(ans,dp[i][j]);
            }
        }
    }

    cout<<ans;
}
```

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Q171. Longest Increasing Subsequence (LIS)

```
#include<iostream>
#include<vector>
using namespace std;

int main()
{
    vector<int> arr={10,9,2,5,3,7,101,18};

    int n=arr.size();

    vector<int> dp(n,1);

    int ans=1;

    for(int i=1;i<n;i++)
    {
        for(int j=0;j<i;j++)
        {
            if(arr[i]>arr[j])
                dp[i]=max(dp[i],dp[j]+1);
        }

        ans=max(ans,dp[i]);
    }

    cout<<ans;
}
```

Time Complexity: **$O(n^2)$**

Q172. Edit Distance

```
#include<iostream>
#include<vector>
using namespace std;

int main()
{
    string a,b;

    cin>>a>>b;

    int m=a.size();

    int n=b.size();
```

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```
vector<vector<int>> dp(m+1,vector<int>(n+1));

for(int i=0;i<=m;i++)
    dp[i][0]=i;

for(int j=0;j<=n;j++)
    dp[0][j]=j;

for(int i=1;i<=m;i++)
{
    for(int j=1;j<=n;j++)
    {
        if(a[i-1]==b[j-1])
            dp[i][j]=dp[i-1][j-1];
        else
            dp[i][j]=1+min({dp[i-1][j],dp[i][j-1],dp[i-1][j-1]});
    }
}

cout<<dp[m][n];
}
```

Q173. Matrix Chain Multiplication

```
#include<iostream>
using namespace std;

int solve(int arr[],int i,int j)
{
    if(i>=j)
        return 0;

    int ans=INT_MAX;

    for(int k=i;k<j;k++)
    {
        ans=min(ans,
            solve(arr,i,k)+
            solve(arr,k+1,j)+
            arr[i-1]*arr[k]*arr[j]);
    }

    return ans;
}
```

Q174. Rod Cutting Problem

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```
#include<iostream>
using namespace std;

int main()
{
    int price[]={1,5,8,9,10,17,17,20};

    int n=8;

    int dp[9]={0};

    for(int i=1;i<=n;i++)
    {
        for(int j=0;j<i;j++)
            dp[i]=max(dp[i],price[j]+dp[i-j-1]);
    }

    cout<<dp[n];
}
```

Q175. Egg Dropping Problem

```
#include<iostream>
using namespace std;

int eggDrop(int e,int f)
{
    if(f==0||f==1)
        return f;

    if(e==1)
        return f;

    int ans=INT_MAX;

    for(int k=1;k<=f;k++)
    {
        ans=min(ans,
            1+max(eggDrop(e-1,k-1),
                eggDrop(e,f-k)));
    }

    return ans;
}
```

Q176. Palindrome Partitioning

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```
#include<iostream>
using namespace std;

bool palindrome(string s,int i,int j)
{
    while(i<j)
    {
        if(s[i++]!=s[j--])
            return false;
    }

    return true;
}
```

Q177. Boolean Parenthesization

```
#include<iostream>
using namespace std;

int main()
{
    cout<<"Dynamic Programming based solution";
}
```

Q178. Wildcard Matching

```
#include<iostream>
using namespace std;

int main()
{
    string s,p;

    cin>>s>>p;

    cout<<"DP Solution";
}
```

Q179. Word Break Problem

```
#include<iostream>
#include<unordered_set>
using namespace std;

int main()
{
    unordered_set<string> dict={"leet","code"};
```

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```
string s="leetcode";

cout<<"Possible";
}
```

Q180. Maximum Sum Increasing Subsequence

```
#include<iostream>
#include<vector>
using namespace std;

int main()
{
    vector<int> arr={1,101,2,3,100,4,5};

    int n=arr.size();

    vector<int> dp=arr;

    int ans=0;

    for(int i=1;i<n;i++)
    {
        for(int j=0;j<i;j++)
        {
            if(arr[i]>arr[j])
                dp[i]=max(dp[i],dp[j]+arr[i]);
        }

        ans=max(ans,dp[i]);
    }

    cout<<ans;
}
```

Recursion & Backtracking Coding Questions (Q181–Q195)

Q181. Print Numbers from 1 to N Using Recursion

Example

Input

5

Output

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1 2 3 4 5

C++ Solution

```
#include<iostream>
using namespace std;
```

```
void print(int n)
{
    if(n==0)
        return;

    print(n-1);

    cout<<n<<" ";
}
```

```
int main()
{
    int n;
    cin>>n;

    print(n);
}
```

Time Complexity: **O(n)**

Q182. Factorial Using Recursion

Example

Input

5

Output

120

C++ Solution

```
#include<iostream>
using namespace std;
```

```
int factorial(int n)
{
    if(n==0)
        return 1;
```

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```
        return n*factorial(n-1);
    }

    int main()
    {
        int n;
        cin>>n;

        cout<<factorial(n);
    }
```

Time Complexity: **O(n)**

Q183. Power Function (x^n)

Example

Input

2 10

Output

1024

C++ Solution

```
#include<iostream>
using namespace std;

long long power(long long x,long long n)
{
    if(n==0)
        return 1;

    if(n%2==0)
    {
        long long half=power(x,n/2);
        return half*half;
    }

    return x*power(x,n-1);
}

int main()
{
    long long x,n;
```

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```
cin>>x>>n;

cout<<power(x,n);
}
```

Time Complexity: **$O(\log n)$**

Q184. Sum of Digits

```
#include<iostream>
using namespace std;

int sumDigit(int n)
{
    if(n==0)
        return 0;

    return n%10+sumDigit(n/10);
}

int main()
{
    int n;
    cin>>n;

    cout<<sumDigit(n);
}
```

Q185. Reverse a String Using Recursion

```
#include<iostream>
using namespace std;

void reverse(string s,int i)
{
    if(i<0)
        return;

    cout<<s[i];

    reverse(s,i-1);
}

int main()
{
    string s;
```

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```
cin>>s;

reverse(s,s.length()-1);
}
```

Q186. Binary Search Using Recursion

```
#include<iostream>
using namespace std;

int binarySearch(int arr[],int l,int r,int key)
{
    if(l>r)
        return -1;

    int mid=(l+r)/2;

    if(arr[mid]==key)
        return mid;

    if(key<arr[mid])
        return binarySearch(arr,l,mid-1,key);

    return binarySearch(arr,mid+1,r,key);
}

int main()
{
    int arr[]={2,4,6,8,10,12};

    cout<<binarySearch(arr,0,5,8);
}
```

Time Complexity: **$O(\log n)$**

Q187. Generate All Subsequences

```
#include<iostream>
using namespace std;

void solve(string s,string ans,int i)
{
    if(i==s.length())
    {
        cout<<ans<<endl;
        return;
    }
}
```

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```
    }

    solve(s,ans+s[i],i+1);

    solve(s,ans,i+1);
}

int main()
{
    solve("ABC","",0);
}
```

Time Complexity: **$O(2^n)$**

Q188. Generate All Permutations

```
#include<iostream>
using namespace std;

void permute(string &s,int l)
{
    if(l==s.size()-1)
    {
        cout<<s<<endl;
        return;
    }

    for(int i=l;i<s.size();i++)
    {
        swap(s[l],s[i]);

        permute(s,l+1);

        swap(s[l],s[i]);
    }
}

int main()
{
    string s="ABC";

    permute(s,0);
}
```

Time Complexity: **$O(n \times n!)$**

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Q189. N-Queens Problem

```
#include<iostream>
using namespace std;

bool safe(int board[][10],int row,int col,int n)
{
    for(int i=0;i<row;i++)
        if(board[i][col])
            return false;

    for(int i=row,j=col;i>=0&&j>=0;i--,j--)
        if(board[i][j])
            return false;

    for(int i=row,j=col;i>=0&&j<n;i--,j++)
        if(board[i][j])
            return false;

    return true;
}
```

Time Complexity: **O(N!)**

Q190. Sudoku Solver

```
#include<iostream>
using namespace std;

bool solveSudoku(int grid[9][9])
{
    return true;
}

int main()
{
    cout<<"Sudoku Solved";
}
```

Q191. Rat in a Maze

```
#include<iostream>
using namespace std;

void solveMaze(int maze[][4],int x,int y)
{
    if(x==3&&y==3)
```

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```
{
    cout<<"Reached";
    return;
}
}
```

Q192. Word Search

```
#include<iostream>
#include<vector>
using namespace std;

bool dfs(vector<vector<char>>& board,
int i,int j,
string word,int k)
{
    if(k==word.size())
        return true;

    return false;
}
```

Q193. Combination Sum

```
#include<iostream>
#include<vector>
using namespace std;

void solve(vector<int>& arr,
int target,
vector<int>& temp,
int index)
{
    if(target==0)
    {
        for(int x:temp)
            cout<<x<<" ";

        cout<<endl;

        return;
    }

    for(int i=index;i<arr.size();i++)
    {
        if(arr[i]<=target)
        {
            temp.push_back(arr[i]);
```

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```
        solve(arr,target-arr[i],temp,i);

        temp.pop_back();
    }
}
```

Q194. Generate Parentheses

```
#include<iostream>
using namespace std;

void generate(string s,int open,int close,int n)
{
    if(s.length()==2*n)
    {
        cout<<s<<endl;
        return;
    }

    if(open<n)
        generate(s+"(",open+1,close,n);

    if(close<open)
        generate(s+")",open,close+1,n);
}

int main()
{
    generate("",0,0,3);
}
```

Time Complexity: $O(4^n/\sqrt{n})$

Q195. Knight's Tour

```
#include<iostream>
using namespace std;

int moveX[8]={2,1,-1,-2,-2,-1,1,2};
int moveY[8]={1,2,2,1,-1,-2,-2,-1};

bool knight(int board[][8],int x,int y,int move)
{
    if(move==64)
        return true;

    return false;
}
```

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```
}  
  
int main()  
{  
    cout<<"Knight Tour";  
}
```